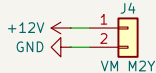
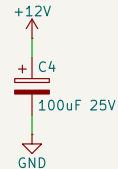
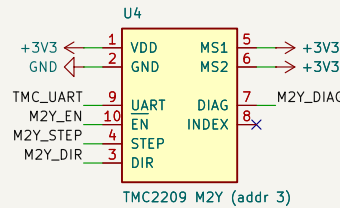
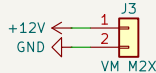
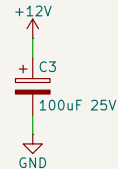
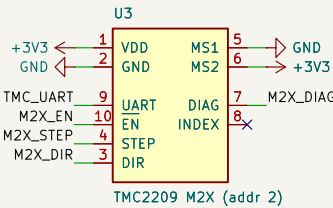
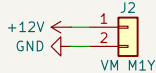
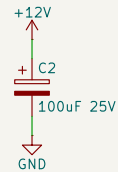
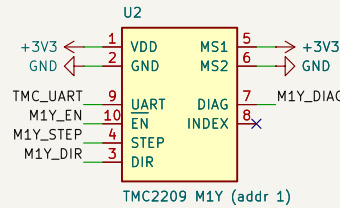
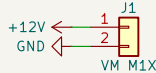
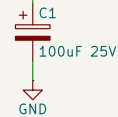
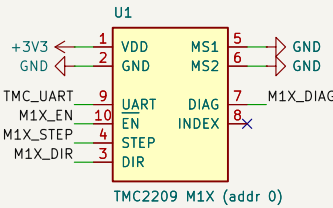
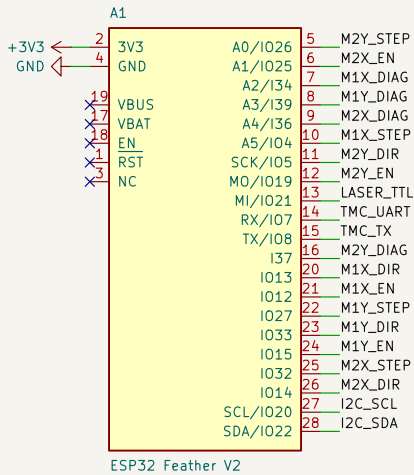


Carrier for the ESP32 Feather V2, four TMC2209 breakouts and the ADS1115.
Pins match firmware/optics_bench/config.h. Wiring: electronics/wiring.md.

Driver UART addresses from MS1/MS2: U1 0, U2 1, U3 2, U4 3.
The breakouts keep their screw terminals: motor wires straight into them, and J1–J4 feed each breakout's VM +/- terminals over a short 2-wire lead.

TX reaches the shared single-wire UART bus through R1; RX joins it directly.
DIAG outputs go to input-only pins I34/I39/I36/I37 (firmware does not use them yet).



12 V motor supply: PTC fuse, series Schottky (reverse polarity), TVS, bulk capacitor

Off-board: laser, photodiode amp boards (1:1 JST–XH cables), test header

